

MIRZA TAIRIN TARIQUE

✉ mirza.tairin@gmail.com

in mirzatairin

☎ +880 1778 722 691

🏠 Dhaka, Bangladesh

ABOUT ME

Computer Science graduate and researcher with interests in Human-centered & Trustworthy AI.

EDUCATION

BRAC University Master of Science in Computer Science and Engineering	Fall 2026–
Ahsanullah University of Science & Technology Bachelor of Science in Computer Science & Engineering	2015 - 2019

WORK EXPERIENCE

Software Developer Global Spirit Engineering Trade & Services Ltd	2023 - 2025
Database Developer East Bengal Transport Service	2021 - 2022
Programmer, MIS Panjeree Publications Ltd	2019 - 2020

PUBLICATIONS

- The Good, the Bad, and the Potential of AI-based Systems in Computing Education
Accepted Paper in IEEE COMPSAC 2025: The ethical implications and human-centered perspectives of AI-assisted learning, as well as the challenges and opportunities of integrating AI-based systems into computer science higher education.

TECHNICAL SKILLS

Programming & Engineering	Python, SQL, Java, C++, Django, .NET, Git
AI & Machine Learning	PyTorch, Scikit-learn, Pandas, NumPy, Computer Vision, Machine Learning, Fairlearn, LLM Fine-Tuning (QLoRA), Hugging Face
Research & Visualization	Matplotlib, Jupyter, ATLAS.ti, StreamLit

RESEARCH PROJECTS

- Gemma for Art (Fine-tuning Multimodal Vision-Language System for Artwork Analysis): Fine-tuned a multimodal vision-language model for art-historical reasoning, contextual interpretation, and structured analysis of visual artworks using QLoRA. (*Google DeepMind Hackathon, 2026*)
- StormCore-3D: Thermodynamic Diagnostic Tool: Multi-field visual analytics framework for tracking and analyzing tropical cyclones across multi-terabyte, multi-resolution planetary simulation (ex. NASA's DYAMOND GEOS dataset). (*IEEE SciVis, 2026*)
- Automated License Plate Recognition (ALPR) System: End-to-end computer vision system for object detection and recognition under real-world environmental variability using a fine-tuned YOLOv8 pipeline (*Undergraduate Thesis*)
- Air Quality Analysis and AQI Modeling: Analyzed urban air quality dynamics using OpenAQ data for environmental modeling and data-driven decision-making. (*AWS AI/ML Scholars Program*)
- Studying Reasoning, Abstraction, and Generalization Across Humans and Intelligent Systems: Investigating reasoning, abstraction, hypothesis generation, and generalization across humans, large language models, and algorithmic systems using ARC as a research environment. (*ARC AGI-3 Paper Track, Ongoing*)

- Algorithmic Bias Mitigation in ML Models: Investigated fairness and bias mitigation in machine learning models, focusing on model reliability, decision transparency, and trustworthy AI.
- Exoplanet Atmosphere Spectral Analyzer: Computational framework simulating and analyzing near-infrared exoplanet atmospheric spectra (*Planetary Science Project*)
- Digital Minds: Human-AI Coexistence Under Moral Uncertainty: A conditional policy framework designed for human–AI interaction under moral uncertainty. (*Apart Research, Submitted for Review*)
- A Minimal Computing Approach to Community-Driven Preservation of Intangible Musical Heritage. (*Research Ireland CFP, Ongoing*)
- Open Source Contributions: SHAP (SHapley Additive exPlanations): Contributed to an open-source explainable AI library for interpreting and auditing machine learning models.

TRAINING/CERTIFICATION

Future AWS AI Programmer Nanodegree Program	2026-ongoing
AWS AI Practitioner Challenge	2026
Introduction to User Experience Design (Georgia Institute of Technology)	2025

INTEREST & HOBBIES

Digital Music Production, Creative Writing, Art, Astrophysics.

LANGUAGE

Bangla	Native
English	IELTS 8.0
Japanese	Beginner

REFERENCES

Available upon request